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United States Patent Application Publication

20250285860

Kind Code

A1

Publication Date

September 11, 2025

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METHOD FOR MANUFACTURING SEMICONDUCTOR TEMPLATE HAVING GROUP III METAL POLAR SURFACE

Abstract

The present invention pertains to a method for manufacturing a semiconductor template, in which a high-quality, highly crystalline Group III metal polar surface is obtained by performing a substrate bonding process only once by selectively growing $\text{Al}_x\text{Ga}_{1-x}\text{N}$ materials having different polarities on a nominal on-axis c-plane sapphire substrate or Si(111) initial growth substrate without using an expensive off-angle sapphire substrate.

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Appl. No.: 18/860668

**Filed (or PCT
Filed):** March 13, 2024

PCT No.: PCT/KR2024/003200

Foreign Application Priority Data

KR 10-2023-0033378

Mar. 14, 2023

Publication Classification

Int. Cl.: H01L21/02 (20060101); C30B25/18 (20060101); C30B29/40 (20060101); C30B33/00 (20060101)

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a method of manufacturing a semiconductor template having a group III metal polar surface, and more specifically, to a method of manufacturing a high-quality semiconductor template having a group III metal polar surface, which has excellent crystallinity, using only a one-step substrate bonding process.

BACKGROUND ART

[0002] Unlike SiC semiconductor devices (metal-oxide-semiconductor field effect transistors (MOSFETs) or diodes) designed and manufactured after being grown on the conventional homogeneous single crystalline SiC initial growth substrate (wafer), group III nitride semiconductor devices are designed and manufactured by epitaxially growing a group III nitride semiconductor directly on a heterogeneous large-diameter (6 inches or more) single crystalline sapphire or an Si initial growth substrate due to the absence of low-cost commercial single crystalline GaN or AlN initial growth substrate, and a semiconductor template (initial growth substrate wafer) having a seed layer with a group III nitride metal polar surface has recently been researched and developed to further improve the quality and performance of the group III nitride semiconductor devices.

[0003] FIG. 1 shows a process of manufacturing a semiconductor template having a group III metal polar surface through the conventional two-step substrate bonding process, and FIG. 2 shows a process of manufacturing a semiconductor template having a group III metal polar surface through the conventional one-step substrate bonding process.

[0004] As shown in FIG. 1, in the conventional two-step substrate bonding process, a semiconductor template having a group III nitride metal polar surface is manufactured by growing a metal (Al or Ga) polarity nitride seed layer **12** on an initial growth substrate **11**, bonding a primary support substrate **14** on the seed layer **12** through a first bonding layer **13**, removing the initial growth substrate **11**, bonding a secondary support substrate **16** on a surface from which the initial growth substrate **11** is removed through a second bonding layer **15**, and removing the primary support substrate **14**.

[0005] The corresponding process is a group III nitride epitaxy growth method that has already been widely verified and has an advantage of having high-quality crystallinity and very low surface roughness, making it easy to secure a mirror-like surface, but there are disadvantages such as a low yield due to two-step substrate bonding and high process costs.

[0006] Meanwhile, as shown in FIG. 2, in the conventional one-step substrate bonding process, a semiconductor template having a group III metal polar surface is manufactured by growing a nitrogen polarity nitride seed layer **22** on an initial growth substrate **21**, bonding a support substrate **24** on the seed layer **22** through a bonding layer **23**, and removing the initial growth substrate **21**.

[0007] The corresponding process has an advantage of increasing a yield and reducing process costs due to the one-step substrate bonding, but in reality, there are disadvantages that it is difficult to grow the seed layer **22** having a nitrogen polar surface with high-quality crystallinity and it is difficult to secure a mirror-like surface due to high surface roughness.

[0008] In addition, $\text{Al}_x\text{Ga}_{1-x}\text{N}$ generally strongly tends to grow to physically have a metal polar state during growth. Due to the physical properties, $\text{Al}_x\text{Ga}_{1-x}\text{N}$ has a mixed polar form with mixed

metal polarity despite trying to grow a nitrogen-polar $\text{Al}_x\text{Ga}_{1-x}\text{N}$, and thus a surface roughness issue, such as hillocks, occurs on a surface.

[0009] To solve the problem of increased surface roughness due to the occurrence of hillocks and the like, there are a method of changing an off angle of an initial growth substrate and a method of adopting a chemical-mechanical polishing (CMP) process using the easily etched property of a non-polar surface, but when an off angle substrate is used or the CMP process is introduced, there is a problem that process costs significantly increase.

DISCLOSURE

Technical Problem

[0010] The present invention is intended to solve the above-described conventional problems and is directed to providing a method of manufacturing a high-quality semiconductor template having a group III metal polar surface, which has excellent crystallinity, using only a one-step substrate bonding process.

Technical Solution

[0011] According to the present invention, the above object is achieved by a method of manufacturing a semiconductor template having a group III metal polar surface, which includes a first growing operation of growing a first semiconductor layer having a metal polar surface on a growth substrate, a layer forming operation of forming a polarity inversion layer on the first semiconductor layer, a second growing operation of growing a second semiconductor layer having a nitrogen polar surface on the polarity inversion layer, a bonding operation of bonding the second semiconductor layer and a support substrate, a separating operation of separating the growth substrate, and a removing operation of removing the first semiconductor layer and the polarity inversion layer to expose a metal polar surface of the second semiconductor layer.

[0012] In addition, the method of the present invention may further include a re-growing operation of re-growing a device layer on the second semiconductor layer of which the metal polar surface is exposed.

[0013] In addition, each of the first semiconductor layer and the second semiconductor layer may be made of $\text{Al}_x\text{Ga}_{1-x}\text{N}$.

[0014] In addition, the polarity inversion layer may be made of a compound including magnesium (Mg) and nitrogen (N).

[0015] According to the present invention, the above object is achieved by a method of manufacturing a semiconductor template having a group III metal polar surface, which includes a first growing operation of growing a first semiconductor layer having a metal polar surface on a growth substrate, a first layer forming operation of forming a first polarity inversion layer on the first semiconductor layer, a second growing operation of growing a second semiconductor layer having a nitrogen polar surface on the first polarity inversion layer, a second layer forming operation of forming a second polarity inversion layer on the second semiconductor layer, a third growing operation of growing a third semiconductor layer having a metal polar surface on the second polarity inversion layer, a bonding operation of bonding the third semiconductor layer and a support substrate, a separating operation of separating the growth substrate, and a removing operation of removing the first semiconductor layer and the first polarity inversion layer to expose a metal polar surface of the second semiconductor layer.

[0016] In addition, the method of the present invention may further include a re-growing operation of re-growing a device layer on the second semiconductor layer of which the metal polar surface is exposed.

[0017] In addition, each of the first semiconductor layer, the second semiconductor layer, and the third semiconductor layer may be made of $\text{Al}_x\text{Ga}_{1-x}\text{N}$.

[0018] In addition, the first polarity inversion layer may be made of a compound including magnesium (Mg) and nitrogen (N), and the second polarity inversion layer may be made of a compound including aluminum (Al) and nitrogen (N).

Advantageous Effects

[0019] According to the present invention, a high-quality semiconductor template having a group III metal polar surface, which has excellent crystallinity, using only a one-step substrate bonding process, can be manufactured by selectively growing $\text{Al}_x\text{Ga}_{1-x}\text{N}$ materials having different polarities on a nominal on-axis c-plane sapphire substrate or an Si(111) initial growth substrate without using an expensive off angle sapphire substrate.

[0020] Meanwhile, the effects of the present invention are not limited to the above-described effects, and may include various effects within a range that is apparent to those skilled in the art from the following descriptions.

Description

DESCRIPTION OF DRAWINGS

[0021] FIG. 1 shows a process of manufacturing a semiconductor template having a group III metal polar surface through the conventional two-step substrate bonding process.

[0022] FIG. 2 shows a process of manufacturing a semiconductor template having a group III metal polar surface through the conventional one-step substrate bonding process.

[0023] FIG. 3 is a flowchart of a method of manufacturing a semiconductor template having a group III metal polar surface according to a first embodiment of the present invention.

[0024] FIG. 4 shows a process of manufacturing a semiconductor template according to the method of manufacturing a semiconductor template having a group III metal polar surface according to the first embodiment of the present invention.

[0025] FIG. 5 is a flowchart of a method of manufacturing a semiconductor template having a group III metal polar surface according to a second embodiment of the present invention.

[0026] FIG. 6 shows a process of manufacturing a semiconductor template according to the method of manufacturing a semiconductor template having a group III metal polar surface according to the second embodiment of the present invention.

MODES OF THE INVENTION

[0027] Hereinafter, some embodiments of the present invention will be described in detail with reference to exemplary drawings. In adding reference numerals to components in each drawing, it should be noted that the same components have the same reference numerals as much as possible even when they are illustrated in different drawings.

[0028] In addition, in describing embodiments of the present invention, detailed descriptions of related known configurations or functions will be omitted when it is determined that the detailed descriptions obscure the understanding of the embodiments of the present invention.

[0029] In addition, terms such as first, second, A, B, (a), and (b) may be used to describe components of the embodiments of the present invention. These terms are only for the purpose of distinguishing one component from another component, and the nature, sequence, order, or the like of the corresponding component is not limited by the terms.

[0030] The present invention relates to a method of manufacturing a high-quality semiconductor template having a group III metal polar surface, which has excellent crystallinity, using only a one-step substrate bonding process by selectively growing $\text{Al}_x\text{Ga}_{1-x}\text{N}$ materials having different polarities on a nominal on-axis c-plane sapphire substrate or an Si initial growth substrate without using an expensive off angle sapphire substrate.

[0031] Hereinafter, a method **S100** of manufacturing a semiconductor template having a group III metal polar surface according to the first embodiment of the present invention will be described in detail with reference to the accompanying drawings.

[0032] FIG. 3 is a flowchart of a method of manufacturing a semiconductor template having a group III metal polar surface according to a first embodiment of the present invention, and FIG. 4

shows a process of manufacturing a semiconductor template according to the method of manufacturing a semiconductor template having a group III metal polar surface according to the first embodiment of the present invention.

[0033] As shown in FIGS. 3 and 4, the method **S100** of manufacturing a semiconductor template having a group III metal polar surface according to the first embodiment of the present invention includes a first growing operation **S110**, a layer forming operation **S120**, a second growing operation **S130**, a bonding operation **S140**, a separating operation **S150**, a removing operation **S160**, and a re-growing operation **S170**.

[0034] The first growing operation **S110** and the second growing operation **S130** of the present invention are operations of growing AlGa_N having different surface polarities on a heterogeneous initial growth substrate **110**, and first, the first growing operation **S110** is an operation of growing a first semiconductor layer **121** having a group III metal polar surface on the initial growth substrate **110**.

[0035] The initial growth substrate **110** may be provided as a sapphire substrate or an Si substrate (nominal on-axis c-plane), and when the initial growth substrate **110** is removed through a laser lift off (LLO) technique in the separating operation **S150** to be described below, the initial growth substrate **110** is preferably provided as an optically transparent and high-temperature heat-resistant sapphire substrate that can transmit 100% (theoretically) of a laser beam (single wavelength light) without absorption, and the initial growth substrate **110** is also preferably provided as a patterned sapphire substrate (PSS) having a protrusion shape that is regularly or irregularly patterned in various dimensions (size and shape) in microscale or nanoscale in order to minimize crystal defects inside a group III nitride semiconductor thin film grown on the initial growth substrate **110**.

[0036] Meanwhile, when the initial growth substrate **110** is removed through a chemical lift off (CLO) technique in the separating operation **S150** to be described below, the initial growth substrate **110** is preferably provided as an Si substrate that may be removed by wet etching, mechanically polished, and selectively etched, and the Si substrate may be made of Si having a (111) crystal face so that a high-quality group III nitride semiconductor thin film may be grown.

[0037] In the first growing operation **S110**, the first semiconductor layer **121** having a group III metal polar surface may be grown on the initial growth substrate **110**, and in this case, the first semiconductor layer **121** may be made of Al_xGa_{1-x}N and may have a thickness of about 1.5 μm.

[0038] The layer forming operation **S120** is an operation of forming a polarity inversion layer **122** on the first semiconductor layer **121**. The polarity inversion layer **122** inverts a surface polarity of a second semiconductor layer **123** to be grown later to nitrogen polar, may be made of a nitride (Mg_xN₂) containing magnesium (Mg) or a nitride (Mg_xSi_yN₂) containing magnesium (Mg) and silicon (Si) through a continuous subsequent in-situ MOCVD process after the first semiconductor layer **121** is grown, and may have a thickness of about 0.5 μm to obtain a nitrogen polar second semiconductor layer **123** having a mirror-like surface. Meanwhile, in the layer forming operation **S120** of the present invention, the polarity inversion layer **122** may be formed through an ex-situ annealing process under an oxygen (O₂) or nitrogen (N₂) atmosphere.

[0039] The second growing operation **S130** is an operation of growing the second semiconductor layer **123** having a nitrogen polar surface on the polarity inversion layer **122**. That is, in the second growing operation **S130**, the second semiconductor layer **123** may be grown to have a nitrogen polar surface through the polarity inversion layer **122**, and in this case, the second semiconductor layer **123** may be made of Al_xGa_{1-x}N and may have a thickness of about 1.5 μm.

[0040] The bonding operation **S140** is an operation of bonding the nitrogen polar surface of the second semiconductor layer **123** and a support substrate **140** through a bonding layer **130**.

[0041] Here, the support substrate **140** is a substrate that supports the second semiconductor layer **123** and a device layer **150** formed on the second semiconductor layer **123** after undergoing each operation of the method **S100** of manufacturing a semiconductor template having a group III metal polar surface according to the first embodiment of the present invention, and in the present

embodiment, a final support substrate **140** is preferably made of a material that may have high heat dissipation performance (60 W/mk or more), have a coefficient of thermal expansion (CTE) that is the same as or similar to that of the device layer **150**, and have a polycrystalline microstructure as a result of high-temperature sintering processing, and the material may include, for example, Si (149 W/mk, 2.6 ppm), SiC (300 to 450 W/mk, 4.8 ppm), AlN (170 to 230 W/mk, 4.5 ppm), SiNx (90 W/mk, 3.7 ppm), sapphire, etc., but is not limited thereto.

[0042] In the past, epitaxial wafer bowing occurred due to the thermo-mechanical induced stress caused by differences in lattice constant (LC) and CTE between the initial growth substrate **110** and a group III nitride semiconductor, however, in the present invention, this can be resolved by strongly bonding the support substrate **140** to one surface of the second semiconductor layer **123** through the bonding layer **130**. That is, since the epitaxial wafer to which the support substrate **140** is bonded is in a stress-relieved state and wafer bowing can be minimized to almost zero, there is an advantage that this method can be applied to wafers having sizes of not only 4 inches, 6 inches, and 8 inches, but also 12 inches or more.

[0043] Meanwhile, in the bonding operation **S140**, the nitrogen polar surface of the second semiconductor layer **123** may be bonded to the support substrate **140** by forming a first bonding layer **130** on one surface of the second semiconductor layer **123**, forming a second bonding layer **130** on the support substrate **140**, and then bonding the first bonding layer **130** and the second bonding layer **130**.

[0044] Here, the bonding layer **130** (including the first bonding layer **130** and the second bonding layer **130**) may be preferentially selected from dielectric materials whose properties do not change and which have excellent thermal conductivity in the MOCVD chamber (at a temperature of 1000° C. or higher and in a reducing atmosphere) in which the group III nitride semiconductor is grown, and may contain, for example, SiO₂ (0.8 ppm), SiNx (3.7 ppm), SiCN (3.8 to 4.8 ppm), AlN (4.5 ppm), Al₂O₃ (6.8 ppm), or amorphous Si, and furthermore, may contain a flowable oxide (FOX) such as spin on glass (SOG) (liquid SiO₂) or hydrogen silsesquioxane (HSQ) to improve surface roughness. Furthermore, the bonding layer **130** may be made of a metal such as Al, W, or Mo, or an alloy thereof.

[0045] Therefore, in the present invention, the second semiconductor layer **123** grown to high quality is bonded with a high heat-resistant bonding material layer on the support substrate **140** having high heat dissipation performance, thereby enabling the single crystalline growth of a high-quality group III nitride semiconductor at a high temperature of 700° C. or higher.

[0046] Meanwhile, in the bonding operation **S140** of the present invention, a reinforcing layer and the first bonding layer **130** may be sequentially stacked on one surface of the second semiconductor layer **123**, a reinforcing layer and the second bonding layer **130** may be sequentially stacked on the support substrate **140**, and then the first bonding layer **130** and the second bonding layer **130** may be pressed to form the bonding layer **130**.

[0047] Here, the reinforcing layer is a layer for reinforcing the bonding strength with the support substrate **140** and inducing compressive stress, and more specifically, the reinforcing layer may include a bonding reinforcing layer and a compressive stress layer.

[0048] The bonding reinforcing layer is a layer introduced to increase bonding strength when the second semiconductor layer **123** is bonded on the support substrate **140** through the bonding layer **130**, and a material constituting the bonding reinforcing layer is preferably selected from SiO₂, SiNx, etc.

[0049] The compressive stress layer is made of a dielectric material with a greater CTE than the support substrate **140**, for example, a material that relieves tensile stress, that is, induces compressive stress, such as AlN (4.6 ppm), AlNO (4.6 to 6.8 ppm), Al₂O₃ (6.8 ppm), SiC (4.8 ppm), SiCN (3.8 to 4.8 ppm), GaN (5.6 ppm), or GaNO (5.6 to 6.8 ppm), which serves to guide the quality improvement of products through stress control.

[0050] Meanwhile, in the present invention, the bonding reinforcing layer or compressive stress

layer may be omitted in some cases, and in some cases, the entire reinforcing layer may be omitted so that one surface of the second semiconductor layer **123** is in direct contact with the bonding layer **130** or the support substrate **140** is in direct contact with the bonding layer **130**. This case may be a structure that induces compressive stress together with a bonding function by forming the bonding layer **130** using a material having a greater CTE than the support substrate **140**.

[0051] The separating operation **S150** is an operation of separating the initial growth substrate **110** from the first semiconductor layer **121** using an LLO technique or CLO technique depending on the material of the initial growth substrate **110**.

[0052] Here, the LLO technique is a technique used when the initial growth substrate **110** is provided as a sapphire substrate and is a technique of separating an epitaxy-grown layer from the initial growth substrate **110** by irradiating a back surface of the transparent initial growth substrate **110** with an ultraviolet (UV) laser beam having a uniform optical output and beam profile, and a single wavelength. When the initial growth substrate **110** is separated, the insides of the first semiconductor layer **121** and the second semiconductor layer **123** that are transferred onto the support substrate **140** are in a state in which the stress has been completely relieved and maintain a flat state together with the support substrate **140**.

[0053] In addition, the CLO technique is a technique used when the initial growth substrate **110** is provided as an Si substrate and is a technique of separating and removing an Si material of the initial growth substrate **110** through wet etching using a tetramethylammonium hydroxide (TMAH) or hydrofluoric+nitric+acetic acids (HNA) solution to completely remove the remaining thin Si after mechanically grinding and polishing the back surface of the Si initial growth substrate **110** having a (111) crystal face. When the initial growth substrate **110** is separated, the insides of the first semiconductor layer **121** and the second semiconductor layer **123** that are transferred onto the support substrate **140** are in a state in which the stress has been completely relieved and maintain a flat state together with the support substrate **140**. Meanwhile, before removing the residual Si material after mechanically grinding and polishing the initial growth substrate **110**, the initial growth substrate **110** is preferably protected from an etchant by forming a protective film such as SiO₂ or SiN_x on a back surface of the support substrate **140**.

[0054] The removing operation **S160** is an operation of removing the first semiconductor layer **121** and the polarity inversion layer **122** to expose the group III metal polar surface of the second semiconductor layer **123**.

[0055] More specifically, the removing operation **S160** is an operation of completely removing the first semiconductor layer **121** remaining after separating the initial growth substrate **110**, and in this case, since an externally exposed surface of the first semiconductor layer **121** has a nitrogen polarity, the above surface may be easily decomposed through a chemical reaction in a base or basic salt solution, and the polarity inversion layer **122** may also be easily removed through a chemical reaction.

[0056] Then, since a surface of the second semiconductor layer **123** that is exposed externally after the first semiconductor layer **121** and the polarity inversion layer **122** are removed has a metal polarity, the above surface functions as an etching stop film that does not undergo a chemical reaction even in a base or basic salt solution, and a mirror-like surface may be obtained without an additional CMP process.

[0057] The re-growing operation **S170** is an operation of re-growing a high-quality group III nitride-based device layer **150** on the second semiconductor layer **123** of which the group III metal polar surface is exposed.

[0058] That is, in the re-growing operation **S170**, a high-quality group III nitride-based device layer **150** for a power semiconductor device such as an HEMT, a light emitting element such as a light emitting diode (LED), or a communication filter device is re-grown using MOCVD.

[0059] Hereinafter, a method **S200** of manufacturing a semiconductor template having a group III metal polar surface according to the second embodiment of the present invention will be described

in detail with reference to the accompanying drawings.

[0060] FIG. 5 is a flowchart of a method of manufacturing a semiconductor template having a group III metal polar surface according to a second embodiment of the present invention, and FIG. 6 shows a process of manufacturing a semiconductor template according to the method of manufacturing a semiconductor template having a group III metal polar surface according to the second embodiment of the present invention.

[0061] As shown in FIGS. 5 and 6, the method **S200** of manufacturing a semiconductor template having a group III metal polar surface according to the second embodiment of the present invention includes a first growing operation **S210**, a first layer forming operation **S220**, a second growing operation **S230**, a second layer forming operation **S240**, a third growing operation **S250**, a bonding operation **S260**, a separating operation **S270**, a removing operation **S280**, and a re-growing operation **S290**.

[0062] The first growing operation **S210**, the second growing operation **S230**, and the third growing operation **S250** of the present invention are operations of growing AlGa_N having different surface polarities on a heterogeneous initial growth substrate **210**, and first, the first growing operation **S210** is an operation of growing a first semiconductor layer **221** having a group III metal polar surface on the initial growth substrate **210**.

[0063] Since the following description of the first growing operation **S210** is the same as the above-described method of manufacturing a semiconductor template having a group III metal polar surface according to the first embodiment of the present invention, overlapping descriptions thereof will be omitted.

[0064] The first layer forming operation **S220** is an operation of forming a first polarity inversion layer **222** on the first semiconductor layer **221**. The first polarity inversion layer **222** inverts a surface polarity of a second semiconductor layer **223** to be grown later to nitrogen polar, may be made of a nitride (Mg_xN₂) containing magnesium (Mg) or a nitride (Mg_xSi_yN₂) containing magnesium (Mg) and silicon (Si) through a continuous subsequent in-situ MOCVD process after the first semiconductor layer **221** is grown, and may have a thickness of about 0.5 μm to obtain a nitrogen polar second semiconductor layer **223** having a mirror-like surface. Meanwhile, in the first layer forming operation **S220** of the present invention, the first polarity inversion layer **222** may be formed through an ex-situ annealing process under an oxygen (O₂) or nitrogen (N₂) atmosphere.

[0065] The second growing operation **S230** is an operation of growing the second semiconductor layer **223** having a nitrogen polar surface on the first polarity inversion layer **222**. That is, in the second growing operation **S230**, the second semiconductor layer **223** may be grown to have a nitrogen polar surface through the first polarity inversion layer **222**, and in this case, the second semiconductor layer **223** may be made of Al_xGa_{1-x}N and may have a thickness of about 1.5 μm.

[0066] The second layer forming operation **S240** is an operation of forming a second polarity inversion layer **224** on the second semiconductor layer **223**. The second polarity inversion layer **224** inverts a surface polarity of a third semiconductor layer **225** to be grown later to metal polar and may be made of a compound containing aluminum (Al) and nitrogen (N), and specifically, the second polarity inversion layer **224** may be formed by performing preprocessing (trimethylaluminum (TMAI) or triethylaluminum (TEAL) preflow) using an aluminum source through a continuous subsequent in-situ MOCVD process or forming an AlN or AlON interlayer through an ex-situ PVD (sputtering or E-beam).

[0067] The third growing operation **S250** is an operation of growing the third semiconductor layer **225** having a nitrogen polar surface on the second polarity inversion layer **224**. That is, in the third growing operation **S250**, the third semiconductor layer **225** may be grown to have a metal polar surface through the second polarity inversion layer **224**, and in this case, the third semiconductor layer **225** may be made of Al_xGa_{1-x}N and may have a thickness of about 1.5 μm.

[0068] The bonding operation **S260** is an operation of bonding the metal polar surface of the third semiconductor layer **225** and a support substrate **240** through a bonding layer **230**.

[0069] Here, the support substrate **240** is a substrate that supports the third semiconductor layer **225**, the second semiconductor layer **223**, and a device layer **250** formed on the second semiconductor layer **223** after undergoing each operation of the method **S200** of manufacturing a semiconductor template having a group III metal polar surface according to the second embodiment of the present invention, and in the bonding operation **S260**, the metal polar surface of the third semiconductor layer **225** may be bonded to the support substrate **240** by forming a first bonding layer **230** on one surface of the support substrate **240**, forming a second bonding layer **230** on the support substrate **240**, and then bonding the first bonding layer **230** and the second bonding layer **230**. Meanwhile, in the bonding operation **S260**, a reinforcing layer and the first bonding layer **230** may be sequentially stacked on one surface of the third semiconductor layer **225**, a reinforcing layer and the second bonding layer **230** may be sequentially stacked on the support substrate **240**, and then the first bonding layer **230** and the second bonding layer **230** may be pressed to form the bonding layer **230**.

[0070] Since the following description of the bonding operation **S260** is the same as the above-described method of manufacturing a semiconductor template having a group III metal polar surface according to the first embodiment of the present invention, overlapping descriptions thereof will be omitted.

[0071] The separating operation **S270** is an operation of separating the initial growth substrate **210** from the first semiconductor layer **221** using an LLO technique or CLO technique depending on the material of the initial growth substrate **210**.

[0072] Since the following description of the separating operation **S270** is the same as the above-described method of manufacturing a semiconductor template having a group III metal polar surface according to the first embodiment of the present invention, overlapping descriptions thereof will be omitted.

[0073] The removing operation **S280** is an operation of removing the first semiconductor layer **221** and the first polarity inversion layer **222** to expose the metal polar surface of the second semiconductor layer **223**.

[0074] More specifically, the removing operation **S280** is an operation of completely removing the first semiconductor layer **221** remaining after separating the initial growth substrate **210**, and in this case, since an externally exposed surface of the first semiconductor layer **221** has a nitrogen polarity, the above surface may be easily decomposed through a chemical reaction in a base or basic salt solution, and the first polarity inversion layer **222** may also be easily removed through a chemical reaction.

[0075] Then, since a surface of the second semiconductor layer **223** that is exposed externally after the first semiconductor layer **221** and the first polarity inversion layer **222** are removed has a metal polarity, the above surface functions as an etching stop film that does not undergo a chemical reaction even in a base or basic salt solution, and a mirror-like surface may be obtained without an additional CMP process.

[0076] The re-growing operation **S290** is an operation of re-growing a high-quality group III nitride-based device layer **250** on the second semiconductor layer **223** of which the group III metal polar surface is exposed.

[0077] That is, in the re-growing operation **S290**, a high-quality group III nitride-based device layer **250** for a power semiconductor device such as an HEMT, a light emitting element such as a light emitting diode (LED), or a communication filter device is re-grown using MOCVD.

[0078] As described above, although all the components constituting embodiments of the present invention are described as being combined or combined to operate as one, the present invention is not necessarily limited to these embodiments. That is, one or more of all the components may be combined to operate as one without departing from the scope of the purpose of the present invention.

[0079] In addition, the term such as “comprise,” “constitute,” or “have” described above means

that the corresponding component can be included unless otherwise stated, and thus should be construed as further including another component rather than excluding another component. All terms including technical or scientific terms have the same meaning as commonly understood by those skilled in the art to which the present invention pertains unless defined otherwise. Commonly used terms, such as terms defined in a dictionaries, should be interpreted as being consistent with the contextual meaning of the related art and are not interpreted in an ideal or excessively formal meaning unless explicitly defined herein.

[0080] In addition, the above description is merely an exemplary description of the technical spirit of the present invention, and those skilled in the art to which the present invention pertains will be able to variously modify and change the present invention without departing from the essential characteristics of the present invention.

[0081] Therefore, embodiments disclosed in the present invention are not intended to limit the technical spirit of the present invention, but intended to describe the same, and the scope of the technical spirit of the present invention is not limited by these embodiments. The scope of the present invention should be construed by the appended claims, and all technical ideas within the equivalent scope should be construed as being included in the scope of the present invention.

Claims

1. A method of manufacturing a semiconductor template having a group III metal polar surface, the method comprising: a first growing operation of growing a first semiconductor layer having a metal polar surface on a growth substrate; a layer forming operation of forming a polarity inversion layer on the first semiconductor layer; a second growing operation of growing a second semiconductor layer having a nitrogen polar surface on the polarity inversion layer; a bonding operation of bonding the second semiconductor layer and a support substrate; a separating operation of separating the growth substrate; and a removing operation of removing the first semiconductor layer and the polarity inversion layer to expose a metal polar surface of the second semiconductor layer.
2. The method of claim 1, further comprising a re-growing operation of re-growing a device layer on the second semiconductor layer of which the metal polar surface is exposed.
3. The method of claim 1, wherein each of the first semiconductor layer and the second semiconductor layer is made of $\text{Al}_{\text{sub}}\text{xGa}_{\text{sub}}1-\text{xN}$.
4. The method of claim 1, wherein the polarity inversion layer is made of a compound including magnesium (Mg) and nitrogen (N).
5. A method of manufacturing a semiconductor template having a group III metal polar surface, the method comprising: a first growing operation of growing a first semiconductor layer having a metal polar surface on a growth substrate; a first layer forming operation of forming a first polarity inversion layer on the first semiconductor layer; a second growing operation of growing a second semiconductor layer having a nitrogen polar surface on the first polarity inversion layer; a second layer forming operation of forming a second polarity inversion layer on the second semiconductor layer; a third growing operation of growing a third semiconductor layer having a metal polar surface on the second polarity inversion layer; a bonding operation of bonding the third semiconductor layer and a support substrate; a separating operation of separating the growth substrate; and a removing operation of removing the first semiconductor layer and the first polarity inversion layer to expose a metal polar surface of the second semiconductor layer.
6. The method of claim 5, further comprising a re-growing operation of re-growing a device layer on the second semiconductor layer of which the metal polar surface is exposed.
7. The method of claim 5, wherein each of the first semiconductor layer, the second semiconductor layer, and the third semiconductor layer is made of $\text{Al}_{\text{sub}}\text{xGa}_{\text{sub}}1-\text{xN}$.
8. The method of claim 5, wherein the first polarity inversion layer is made of a compound

including magnesium (Mg) and nitrogen (N), and the second polarity inversion layer is made of a compound including aluminum (Al) and nitrogen (N).
